

Logistics & Supply Chain Performance Dashboard

A Power BI project analyzing order fulfillment, warehouse efficiency, and sales performance across the United States.

US Regional Sales Data · Star-Schema Model · 12 DAX Measures

Project Overview

Objective

Build an interactive Power BI report that evaluates how efficiently orders move through the supply chain — from procurement to final delivery — and highlights which warehouses, sales teams, and products drive performance.

Scope

- Single-country dataset (United States)
- 3-year historical window (2018–2020)
- Order-level transactions with full date lineage
- Star-schema model built for scalability

Key Questions This Report Answers

-  How is revenue trending across regions and time?
-  How long does fulfillment take, from procurement to delivery?
-  Which warehouses deliver fastest and generate most revenue?
-  Which sales teams and products perform best?

Data Source

US Regional Sales Data (Kaggle) — a clean, order-level transactional dataset covering procurement, ordering, shipping, and delivery dates for sales made across the United States.

7,991

Order Records

16

Source Columns

1

Country (USA)

4

Date Fields per Order

Data Structure



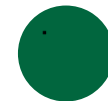
1 Fact Table

Sales — one row per order line



6 Dimension Tables

Warehouse, Sales Team, Customer, Store,
Product, Date



Date Range

January 2018 – December 2020

Fact Table — Sales

Column	Description
OrderNumber	Unique identifier for each order (primary key)
Sales Channel	How the order was placed (In-Store, Online, Distributor, Wholesale)
WarehouseCode	Warehouse that processed the order (→ Dim_Warehouse)
ProcuredDate	Date the ordered products were sourced
OrderDate	Date the customer placed the order
ShipDate	Date the order left the warehouse
DeliveryDate	Date the order reached the customer
CurrencyCode	Currency of the transaction (USD)

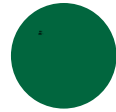
Column	Description
SalesTeamID	Sales team handling the order (→ Dim_SalesTeam)
CustomerID	Customer who placed the order (→ Dim_Customer)
StoreID	Store associated with the order (→ Dim_Store)
ProductID	Product ordered (→ Dim_Product)
Order Quantity	Number of units ordered
Discount Applied	Discount rate applied to the order
Unit Cost	Cost per unit to the company
Unit Price	Price per unit charged to the customer

Dimension Tables



Dim_Warehouse

Warehouse code, name, city, state, region and storage capacity



Dim_SalesTeam

Sales team ID, name, covering region, manager and team size



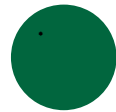
Dim_Customer

Customer ID, name, segment, location and customer-since date



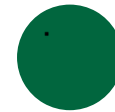
Dim_Store

Store ID, name, store type, location and store-open date



Dim_Product

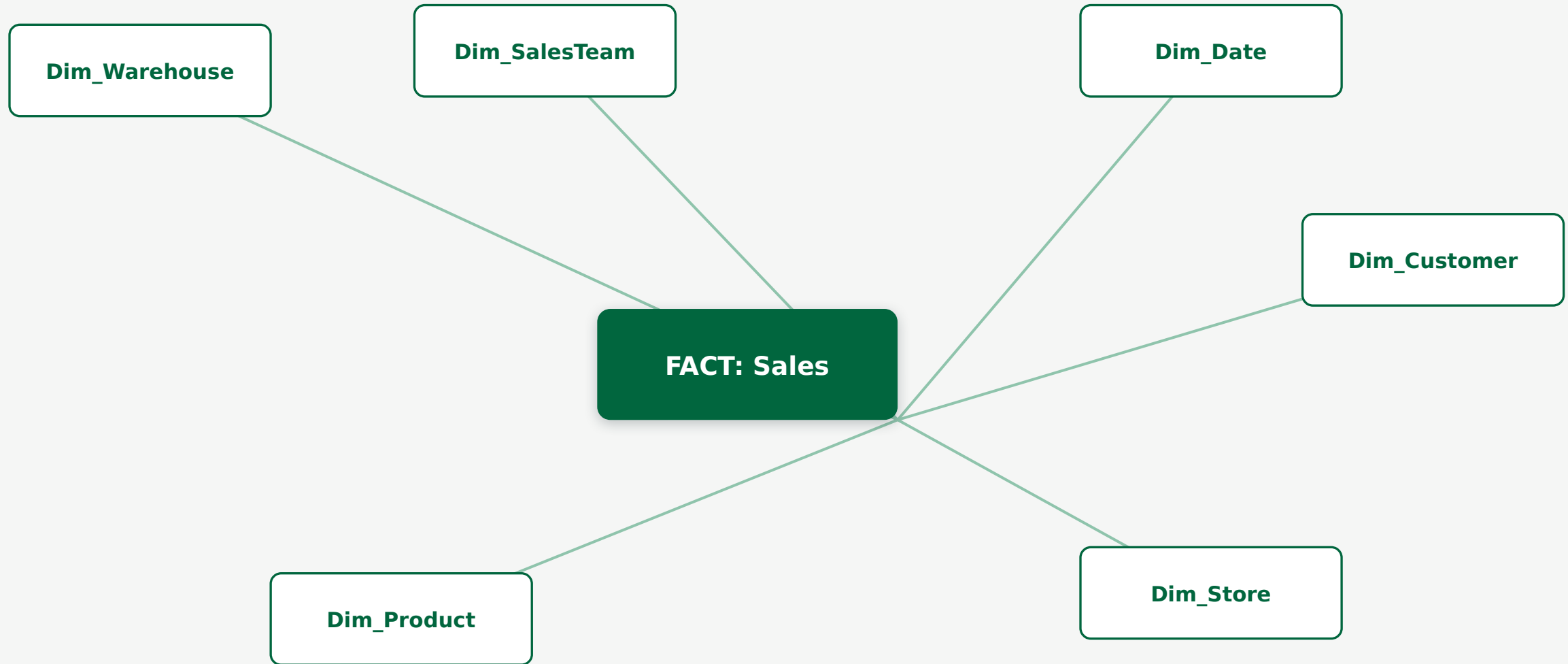
Product ID, name, category, sub-category and unit of measure



Dim_Date

Full calendar table — year, quarter, month, weekday, week number

● Data Model — Star Schema



Each dimension connects to Sales through its ID column. OrderDate drives the active relationship to Dim_Date; ProcuredDate, ShipDate and DeliveryDate use USERELATIONSHIP() for fulfillment-speed measures.

Report Pages

01

Overview

Business performance at a glance



02

Fulfillment Speed

Order-to-delivery timing



03

Warehouse Performance

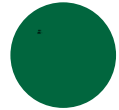
Distribution hub efficiency



04

Sales Team & Product

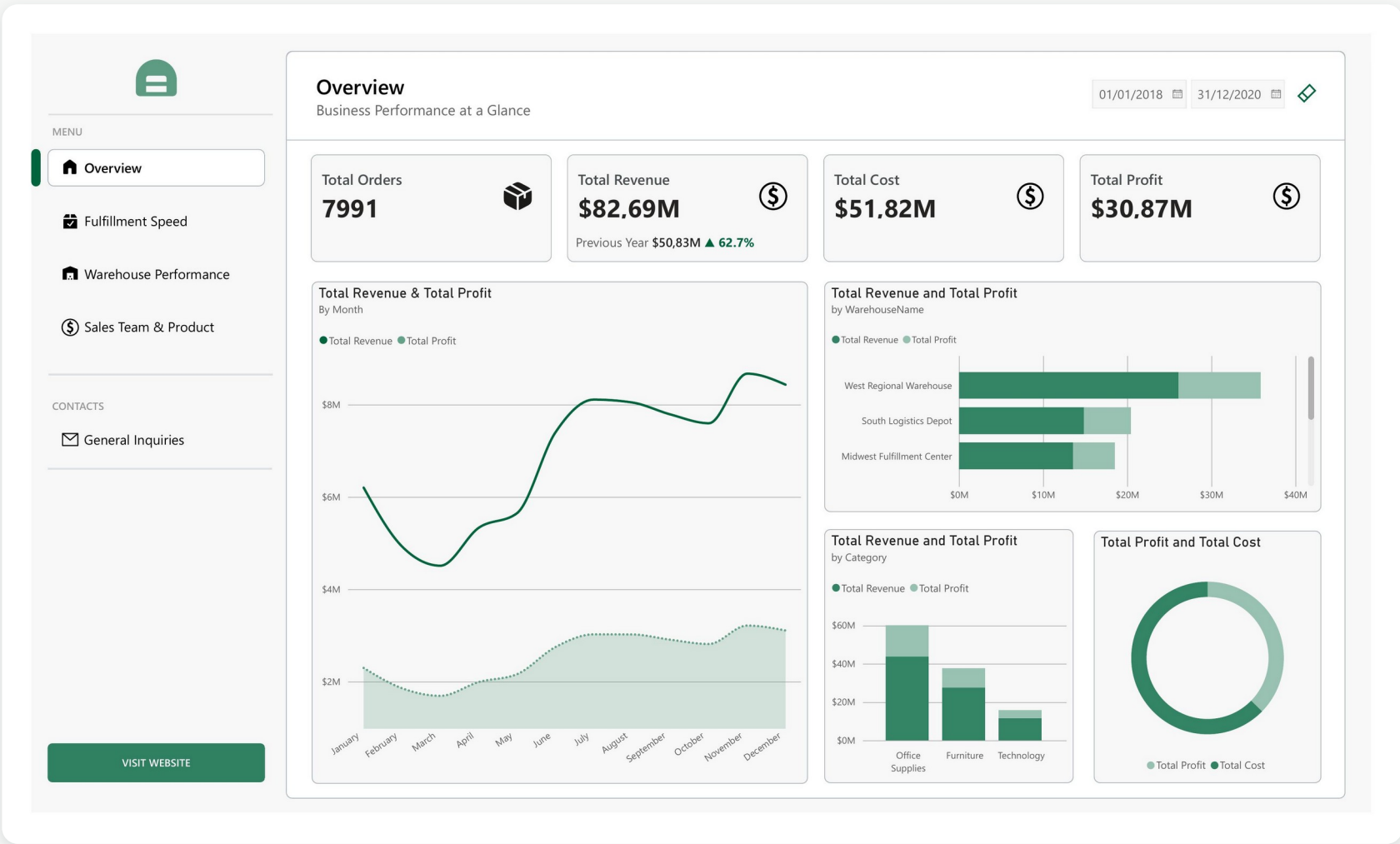
Team and product performance





Page 1 — Overview

Business performance at a glance



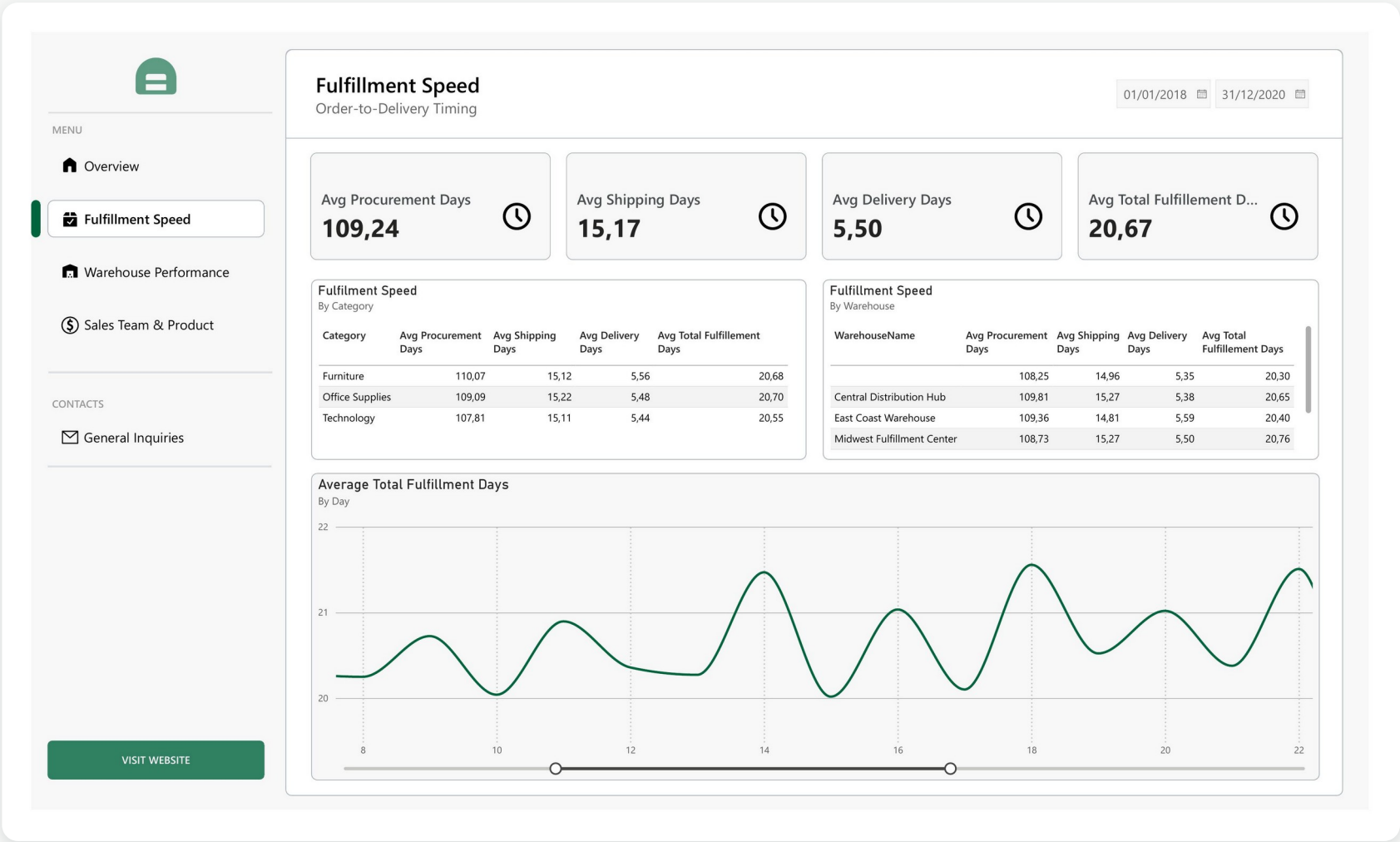
What's on this page

- KPI cards: Total Orders, Revenue, Cost, Profit
- Revenue & Profit trend by month
- Revenue & Profit by Warehouse and Category
- Profit vs. Cost breakdown



Page 2 — Fulfillment Speed

Order-to-delivery timing

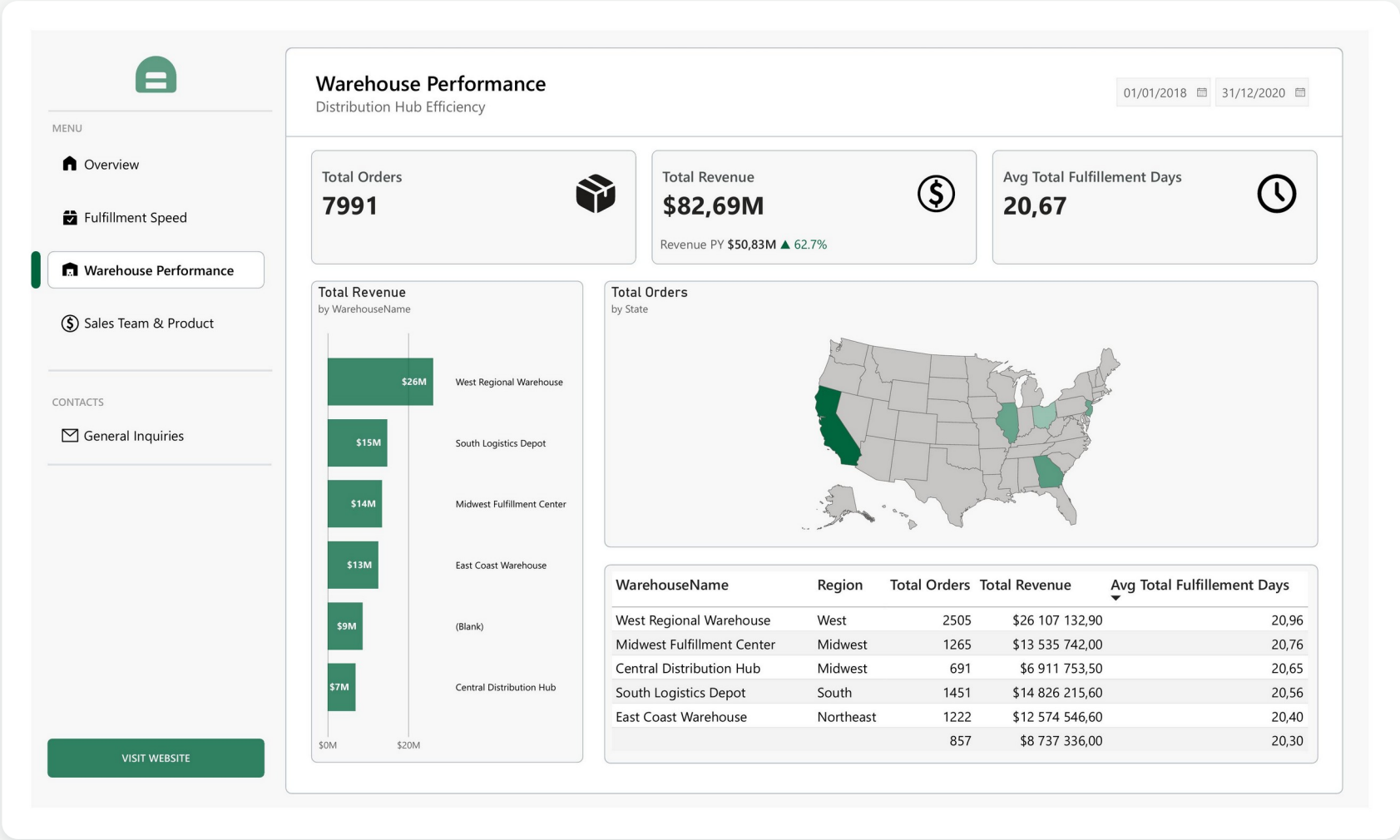


What's on this page

- KPI cards: Avg Procurement, Shipping, Delivery, Total days
- Fulfillment speed by Category
- Fulfillment speed by Warehouse
- Daily trend of total fulfillment time

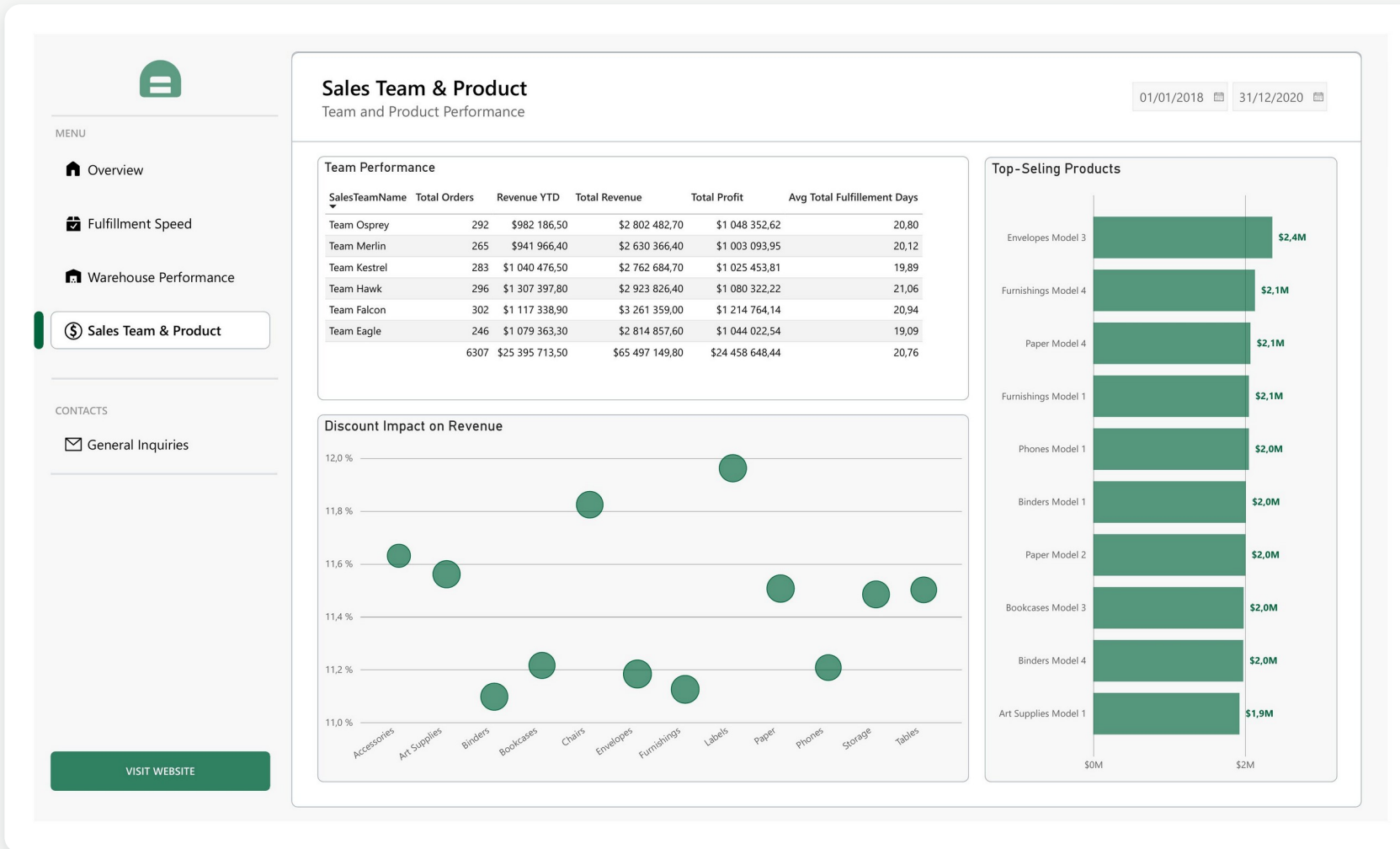
Page 3 — Warehouse Performance

Distribution hub efficiency



Page 4 — Sales Team & Product

Team and product performance



What's on this page

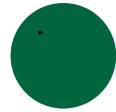
- Team performance table (orders, revenue, profit)
- Discount impact on revenue by category
- Top-selling products, ranked by revenue

Key DAX Measures



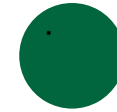
Sales & Revenue

- Total Orders
- Total Revenue
- Total Cost
- Total Profit
- Avg Discount



Fulfillment Speed

- Avg Procurement Days
- Avg Shipping Days
- Avg Delivery Days
- Avg Total Fulfillment Days



Time Intelligence

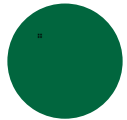
- Revenue YTD
- Revenue PY (Same Period Last Year)
- Revenue Growth %

Deliverables



.pbix Power BI File

Fully built report with 4 interactive pages and applied theme



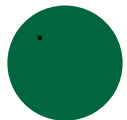
Star-Schema Data Model

1 fact table + 6 dimension tables with defined relationships



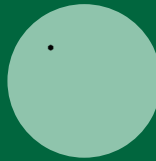
12 DAX Measures

Covering sales, fulfillment speed, and time intelligence



Data Dictionary

Column-level documentation of the fact and dimension tables



Thank You

Logistics & Supply Chain Performance Dashboard — built in Power BI